DATA ANALYSIS PROJECT | HEALTHCARE

Who comes back?

Reducing 30-day readmissions in diabetic hospital patients

1 in 11

first-time diabetic patients is back within 30 days of discharge

Source: 130 US hospitals, 1999-2008 | 101,766 encounters | 50 variables

Problem | Data | Findings | Solution

THE PROBLEM

Patients who come back cost more and signal care that did not stick

A diabetic patient who is discharged and then readmitted within 30 days is an **unplanned return**: it occupies a bed, adds treatment cost, and usually means the transition from hospital to home failed. In this dataset, **11.16%** of all 101,766 hospital stays ended in a 30-day readmission. Counting each patient once (first stay only), the rate is **9.0%**, and **40.7%** of patients come back at some point.



Business question

Follow-up capacity (nurse calls, home visits, clinic slots) is limited. **Which patients should get extra attention at the moment of discharge?**

Goal

Find what drives early readmission, then build a simple risk score that lets the care team focus on the few patients who account for a disproportionate share of returns.

Back within 30 days: 9.0% After 30 days: 31.8% Never: 59.3%

9.0%
of first-time patients return within 30 days

40.7%
return at least once, at any time

29%
readmission rate for patients with 4+ inpatient stays in the prior year

Percentages are based on 69,987 unique patients after cleaning (page 2). The 11.2% figure counts every encounter, so frequent returners are over-represented.

DATA AND PREPARATION

From 101,766 raw rows to a clean, patient-level dataset

The file holds one row per hospital stay with demographics, admission and discharge details, lab tests, 23 diabetes drug columns and the outcome **readmitted** (<30 days, >30 days, or no). It is real, messy hospital data, so cleaning came before any analysis.



Rows kept after each cleaning step

What we did and why

Issue found	Action	Reason
weight missing in 96.9% of rows	Column dropped	Too empty to impute honestly
payer_code 39.6%, **medical_specialty** 49.1%, **race** 2.2% missing	Filled with "Unknown" and kept	Missing itself can be informative; avoids losing half the rows
examide, citoglipton have a single value	Dropped	No information
3 rows with invalid gender	Removed	Cannot be classified
2,423 patients who died or went to hospice	Removed	They cannot be readmitted, which would understate risk
29,353 repeat stays by the same patient	Kept first stay only	Repeat rows are not independent and bias the results
Extreme counts (labs, medications, visits)	Capped at the 99.5th percentile	Limits the pull of a few outliers
Raw ID codes and ICD-9 diagnosis codes	Grouped into readable categories	e.g. discharge to Home / Facility, diagnosis to Circulatory, Respiratory, Diabetes ...

New features created

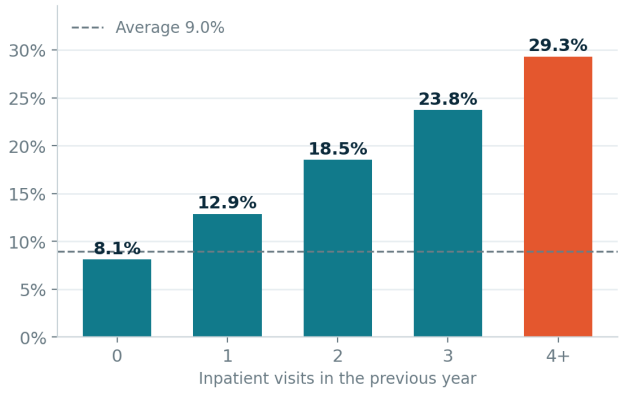
Target readmit_30 (1 if back within 30 days). **Age** as a number and as 6 groups. **Prior visits** (outpatient + ER + inpatient). **Medications used** and **medications changed** during the stay. **Missing lab results** ("None") relabelled "Not tested" so they are not mistaken for missing data.

FINDINGS

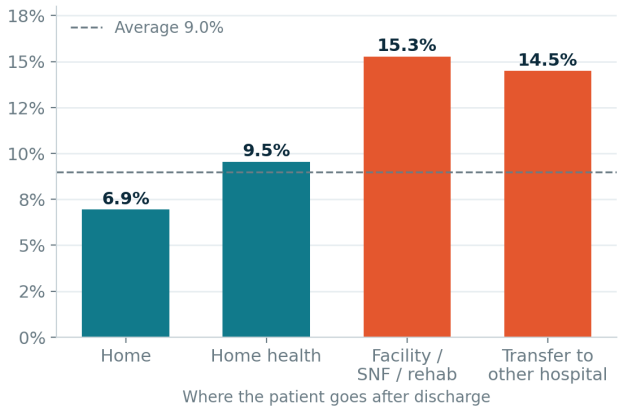
Three signals separate patients who return from those who do not

We compared the 30-day readmission rate across every patient segment. Dashed line = overall average of 9.0%.

1. Prior inpatient stays

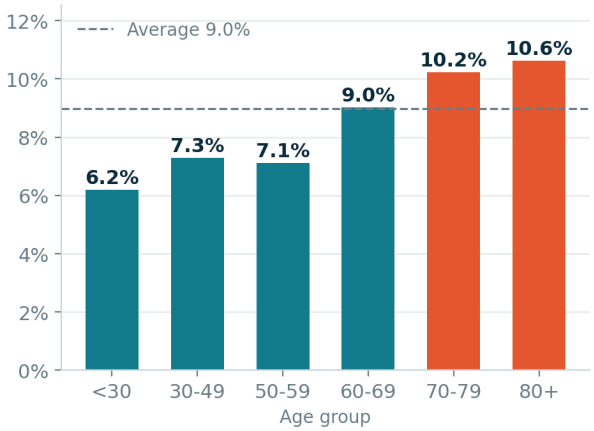


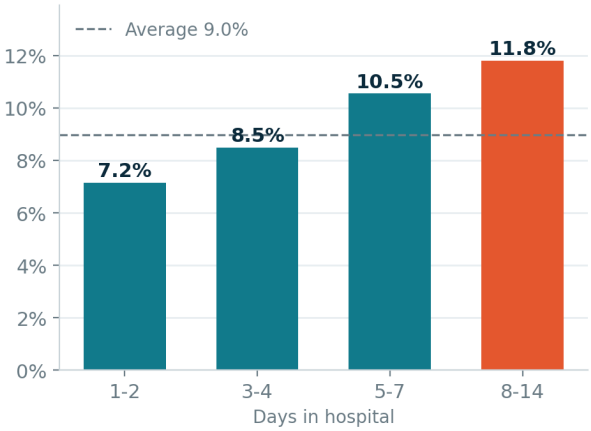
2. Discharge destination



History predicts the future. Risk climbs with every earlier admission: 8.1% with none, 29.3% with four or more, **3.6 times** higher. **Where the patient goes matters.** Discharge to a facility (nursing home, rehab) gives 15.3% versus 6.9% at home, more than double.

3. Age and length of stay





Older patients (70+) and longer stays (8-14 days) run 1 to 3 points above average. These are real but weaker signals than the first two.

2.2x
more risk after a facility discharge than a home discharge

3.6x
more risk with 4+ prior stays than with none

Weak
sex, race and admission type barely move the rate

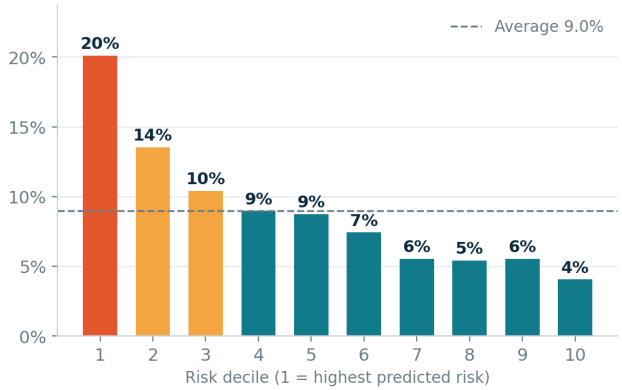
Not a strong driver: A1C testing (8.4% tested vs 9.1% not tested) and medication changes show only small gaps in this data, so they should not be oversold as fixes.

THE SOLUTION

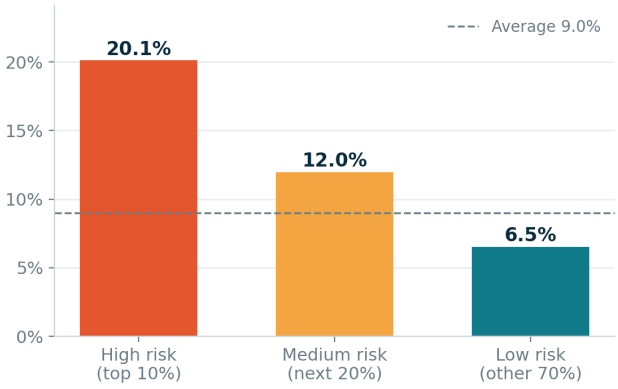
A discharge risk score lets the care team focus where it counts

We trained a gradient-boosting model on 75% of patients and tested it on the other 25% (17,497 unseen patients). Ranking them by predicted risk and cutting into deciles shows the score does sort patients: the top decile returns at **20.1%**, the bottom at **4.1%**.

Readmission rate by risk decile (test set)



Three action tiers



Action plan by tier

Tier	Who	Readmit rate	Share of all returns	Recommended action
High	Top 10% of scores	**20.1%**	**22%**	Discharge planner, follow-up call in 48h, visit within 7 days, medication review
Medium	Next 20%	**12.0%**	**27%**	Follow-up call within a week, written care plan, clinic slot booked before leaving
Low	Other 70%	**6.5%**	**51%**	Standard discharge process

What this could deliver

Targeting the top 30% of patients covers **49.1%** of all readmissions. **Illustrative scenario (an assumption, not a measured result):** per 10,000 discharges there are about 900 readmissions; if extra follow-up for High and Medium patients (3,000 people) cut their returns by 20%, roughly **88 readmissions** a year would be avoided, about one in ten.

Limits to keep in mind

The model is a triage tool, not a crystal ball: accuracy (AUC) is **0.648**, modest but typical for this dataset. Data covers 1999-2008, relationships are associations rather than proven causes, and the 20% effect must be confirmed with a small pilot before scaling.